

Analysis of errors of derived slope and aspect related to DEM data properties

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Abstract

One of the obvious sources of errors in digital terrain analysis (DTA) algorithms is that introduced by raster data structure employed by a digital elevation model (DEM). Because of its regular sample space and orientation, the DTA results often show significant octant ‘bias’, presenting obvious visual and numerical error patterns. Moreover, other DEM data properties may also introduce errors in slope and aspect computation, such as data precision and spatial resolution (i.e. grid interval). This paper reports an investigation on the accuracy of algorithms that derive slope and aspect measures from grid DEM. A quantitative methodology has been developed for objective and data-independent assessment of errors generated from the algorithms that extract surface morphological parameters such as slope and aspect from grid DEM.

The generic approach is to use artificial surfaces that can be described by a mathematical model, thus the ‘true’ output value can be pre-determined to avoid uncertainty caused by uncontrollable data errors. Two mathematical surfaces were generated based on ellipsoid (representing convex slopes) and Gauss synthetic surface (representing complex slopes), and the theoretical ‘true’ value of the slope and aspect at any given point on the surfaces could be computed using mathematical inference. Based on these models, tests were made on the results from a number of algorithms for slope and aspect computation. Analysis has been undertaken to find out the spatial and statistical patterns of error distribution so that the influence of data precision, grid resolution, grid orientation and surface complexity can be quantified.

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1. Introduction

The digital elevation model (DEM) has been utilised as one of the core databases in many GIS application practices. This is because the DEM not only provides the description about three-dimensional surface and data foundation for impressive three-dimensional visualisation of geographical data, but also sets the foundation for deriving other surface morphological parameters such as slope, aspect, curvature, slope profile

and catchment areas. These parameters have been widely utilised in hydrological modelling, soil erosion studies and ecological environment simulation, etc. Among the morphological parameters, slope and aspect have been arguably the most frequently utilised in GIS applications.

For most today’s GIS applications, digital elevation data have usually been provided in a grid data structure, so that the term DEM has been widely regarded as digital elevation grid (Theobald, 1989). Although it has been argued that algorithms that derive slope and aspect are already well developed, the accuracy of the derived parameters is unavoidably influenced by the

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DEM data (Liu, 2002), which is only an approximation to the real-world surface. Numerous studies have been reported on the accuracy analysis of slope and aspect algorithms in relation to DEM data errors (Florinsky, 1998a, b), data precision (Theobald, 1989), grid resolution, or grid cell size (Chang and Tsai, 1991; Garbrecht and Martz, 1994; Hodgson, 1995; Florinsky and Kuryakova, 2000; Tang, 2000) and grid orientation (Jones, 1998). Although the errors caused by data precision, grid resolution and orientation are usually not a concern for the visualisation of three-dimensional surface, they could create significant impact on the derived surface parameters, such as slope and aspect, which are also largely related to the utilised algorithms.

Take the grid orientation as an example; given a slope defined by plane $z = 2x + y + 100$, its calculated slope and aspect are constants of 65.9° and 243.4° , respectively. When a DEM with 10-m grid resolution with a 'standard' orientation (0° , i.e. column direction is N–S direction), the maximum downhill slope algorithm (O'Callaghan and Mark, 1984) derives slope and aspect of 64.7° and 225° , respectively. While the DEM is established with an orientation of 30° , the algorithm derives slope and aspect of 59.1° and 285° , respectively. This accounts for an error level up to $\pm 6.8^\circ$ and $\pm 41.6^\circ$ for derived slope and aspect, respectively.

Studies have also been reported on the suitability of different algorithms on a variety of landscapes, with some significant disagreements on the spatial distribution of errors of derived slope and aspect (Skidmore, 1989; Davis and Dozier, 1990; Chang and Tsai, 1991; Carter, 1992; Florinsky, 1998a, b). For example, Chang and Tsai (1991) stated that the aspect error is greater in flat areas, while slope error is more likely associated with steeper landscape. Carter (1992) on the other hand, concluded that both slope and aspect errors are greater in the flat areas. Florinsky (1998a, b) also reported that high errors of data on local topographic variables are typical for flat areas. Davis and Dozier (1990) found that the slope and aspect errors were concentrated in the areas with significant slope change.

To analyse the relationships between errors of derived slope and aspect with DEM data characteristics such as data precision, grid resolution and grid orientation, we describe a method below that utilises artificial surfaces defined by selected polynomials. Six slope and aspect algorithms were selected and applied on the polynomial surfaces and their results were compared with the 'true values' derived from mathematical inference. Based on the test, the comparison between the selected algorithms was made and the accuracy of the algorithms and their suitability in relation to the nature of surface were analysed.

2. Slope and aspect algorithms

At a given point on a surface $z = f(x, y)$, the slope (S) and aspect (A) is defined as a function of gradients at X and Y (i.e. W–E and N–S) directions, i.e.,

$$S = \arctan \sqrt{f_x^2 + f_y^2}, \quad (1)$$

$$A = 270^\circ + \arctan \left(\frac{f_y}{f_x} \right) - 90^\circ \frac{f_x}{|f_x|}, \quad (2)$$

where f_x and f_y are the gradients at N–S and W–E directions, respectively.

From Eqs. (1) and (2), it is clear that the key for slope and aspect computation is the calculation of f_x and f_y . Using a grid-based DEM, the common approach is to use a moving 3×3 window to derive finite differential or local surface fit polynomial for the calculation (Skidmore, 1989; Florinsky, 1998a, b).

Considering the popularity and the use of different algorithms, we have selected six popular algorithms for test, namely,

- Second-order finite difference (2FD, Fleming and Hoffer, 1979; Zevenbergen and Thorne, 1987; Ritter, 1987),

$$f_x = (z_8 - z_2)/2g, \quad f_y = (z_6 - z_4)/2g. \quad (3)$$

- Third-order finite difference (3FD, Sharpnack and Akin, 1969; Horn, 1981; Wood, 1996),

$$f_x = (z_7 - z_1 + z_8 - z_2 + z_9 - z_3)/6g, \\ f_y = (z_3 - z_1 + z_6 - z_4 + z_9 - z_7)/6g. \quad (4)$$

- Third-order finite difference weighted by reciprocal of squared distance (3FDWRSD, Horn 1981),

$$f_x = (z_7 - z_1 + 2(z_8 - z_2) + z_9 - z_3)/8g, \\ f_y = (z_3 - z_1 + 2(z_6 - z_4) + z_9 - z_7)/8g. \quad (5)$$

- Third-order finite difference weighted by reciprocal of distance (3FDWRD, Unwin, 1981),

$$f_x = (z_7 - z_1 + \sqrt{2}(z_8 - z_2) + z_9 - z_3)/(4 + 2\sqrt{2})g, \\ f_y = (z_3 - z_1 + \sqrt{2}(z_6 - z_4) + z_9 - z_7)/(4 + 2\sqrt{2})g. \quad (6)$$

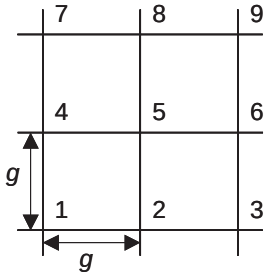
- Frame finite difference (FFD, Chu and Tsai, 1995), and

$$f_x = (z_7 - z_1 + z_9 - z_3)/4g,$$

$$f_y = (z_3 - z_1 + z_9 - z_7)/4g. \quad (7)$$

- Simple difference (SimpleD, Jones, 1998).

$$f_x = (z_5 - z_2)/g, \quad f_y = (z_5 - z_4)/g, \quad (8)$$

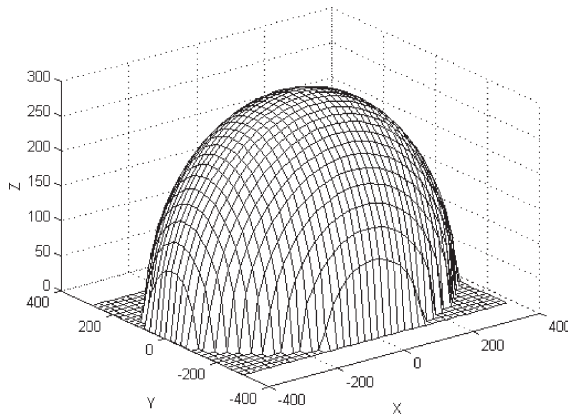
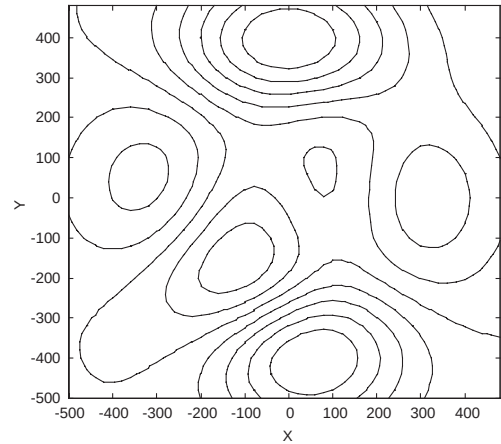
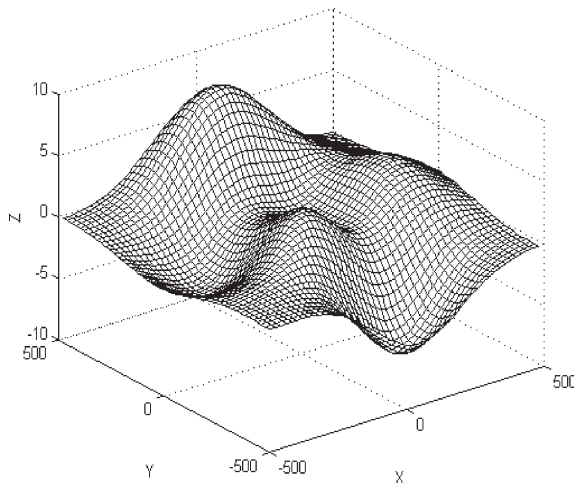
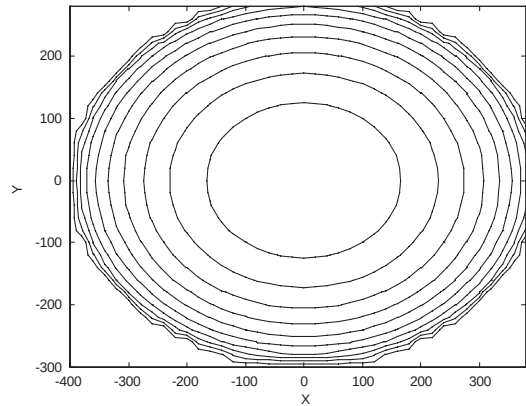
Fig. 1. 3×3 moving window.

where g denotes the grid resolution, while z_i ($1 \leq i \leq 9$) denotes the elevation at each cell of the 3×3 moving window (see Fig. 1). All above equations are for computing f_x and f_y while $i=5$ (i.e. for the centre point).

3. Methodology

In this study, we took the similar approach as reported by Zhou and Liu (2002, 2003) by employing pre-defined standard surfaces for testing and comparing selected algorithms. Our focus is on the influence of data precision, grid resolution and orientation, so that we selected two surfaces for test, namely an ellipsoid (Fig. 2) and a Gauss synthetic surface (Fig. 3), which are defined by the equations below:

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} + \frac{z^2}{C^2} = 1 \quad (z > 0), \quad (9)$$

Fig. 2. Test surface defined by an ellipsoid ($A = 500$, $B = 300$, $C = 300$, $-400 \leq X \leq 400$, $-300 \leq Y \leq 300$, contour interval: 30 m).Fig. 3. Gauss synthetic surface for test ($A = 3$, $B = 10$, $C = 1/3$, $-500 \leq X, Y \leq 500$, contour interval: 1 m).

$$z = A \left[1 - \left(\frac{x}{m} \right)^2 \right] e^{-(x/m)^2 - ((y/n)+1)^2} - B \left[0.2 \left(\frac{x}{m} \right) - \left(\frac{x}{m} \right)^3 - \left(\frac{y}{n} \right)^5 \right] e^{-(x/m)^2 - (y/n)^2} - C e^{-((x/m)+1)^2 - (y/n)^2}, \quad (10)$$

where A , B and C are parameters determining surface relief, and m , n in Eq. (10) are the parameters controlling the spatial extent of the surface. The ‘true values’ of slope and aspect can be computed using the above equations and Eqs. (1) and (2).

The generation of the ‘standard’ surfaces were controlled in terms of their precision, resolution and orientation for the analysis of DEM structure influence. The selected algorithms were then applied on the ‘standard’ surfaces so that the derived slope and aspect can then be compared with the ‘true value’ derived from Eqs. (9) and (10) for each of the grid cell. The root mean square error (RMSE) was then summarised for the results from each algorithm and compared.

4. Results and discussion

4.1. DEM data precision

DEM data precision is indicated by the number of significant digits used for DEM data. In many real-world cases, the DEM precision is defined at the level of 1 m, such as USGS 30 m DEM (Theobald, 1989; Carter, 1992). In some cases, this precision is required to higher levels (e.g. China’s 1:50,000 DEM requires a precision level of 0.1 m, Li and Zhu, 2000).

Usually the DEM error caused by data precision level is quite minimal, except in flat areas where the rounding errors could be significant. As shown in Fig. 4, the

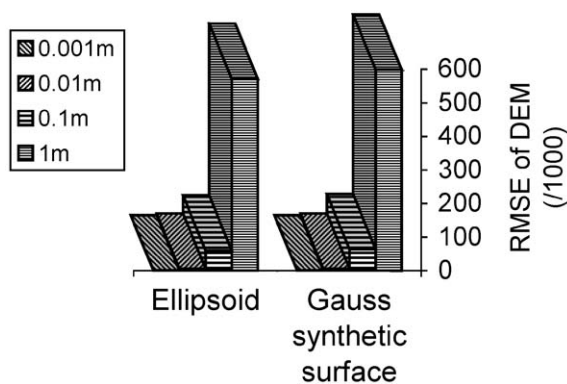


Fig. 4. RMSE of DEM related to different data precision levels.

RMSE of DEM is less than 1 m at 1-m precision level. While the precision level is raised to centimetres, the RMSE of DEM is close to zero.

The influence of data precision on derived slope and aspect is highly related to the grid resolution. While using a high-resolution DEM (e.g. 1 m grid resolution), the influence of data precision becomes quite significant. In order to test the sensitivity of the selected algorithms to data precision, we have created ellipsoid and Gauss synthetic surfaces with 1 m resolution and various precisions at 0.001, 0.01, 0.1 and 1 m levels. The test results on selected algorithms are shown by their RMSE as Fig. 5.

As shown by Fig. 5, when data precision level is reduced from 0.001 to 0.01 m, the change of RMSE of DEM itself is minimal. Further generalisation, however, would cause much more significant increase of RMSE (refer to Fig. 4). For derived slope and aspect, the SimpleD algorithm tends to create much greater RMSE than the others with high precision data. On the other hand, RMSE of all other algorithms seems to increase constantly with decreasing precision, with 2FD showing the most rapid change rate. While reaching the precision level of 1 m, all algorithms show very similar level of RMSE where SimpleD just shows slightly higher error level than the others.

The results show that data precision may only play a significant role in algorithm performance while the precision level is high. When the precision level is reduced, its influence on different algorithms becomes less important. In reality, errors may occur during different stages of DEM generation, such as data capture, sampling and interpolation. Compared to these errors, the rounding errors by reducing data precision can be neglected (with an exception of flat areas). In this case, the number of significant digits should not be considered as critical. When DEM data accuracy is higher than the precision, on the other hand, the data precision error must be considered while selecting algorithms.

4.2. Grid resolution

DEM resolution determines the level of details of the surface being described. It naturally influences the accuracy of derived surface parameters. Numerous studies have been reported on the influence of DEM resolution in relation to a variety of geographical environment using different methods (e.g. Chang and Tsai, 1991; Carter, 1992; Moore et al., 1993; Brasington and Richards, 1998; Gao, 1998; Florinsky, 1998a, b; Tang, 2000). In this study, we focus on two questions:

- Does a high-resolution DEM lead to more accurate estimation of slope and aspect?

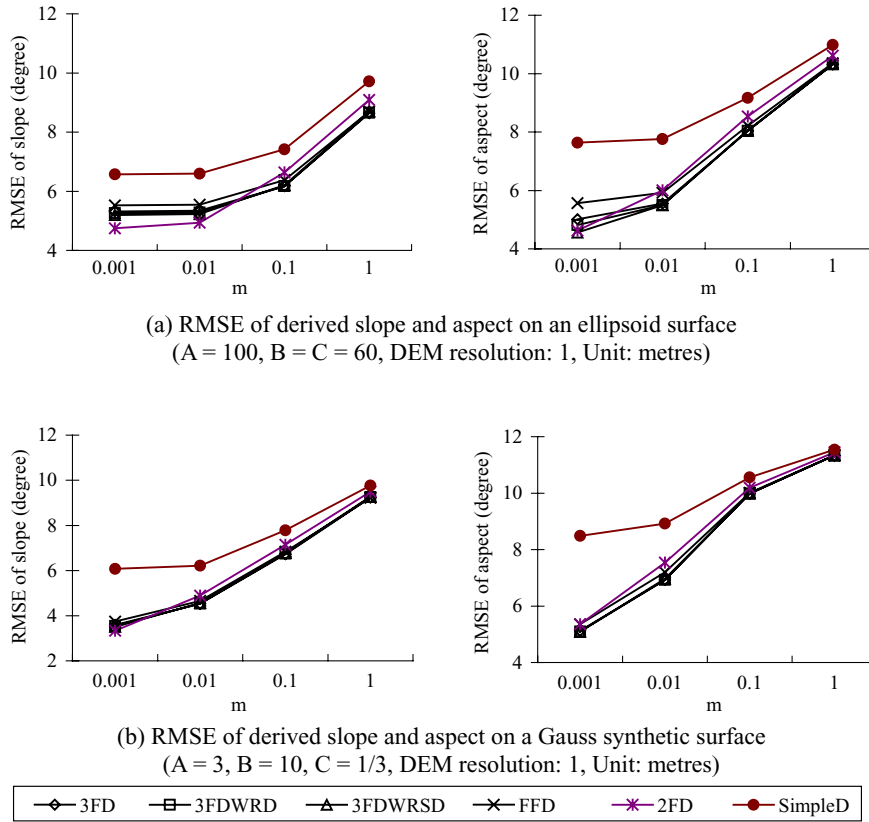


Fig. 5. Influence of DEM data precision on derived slope and aspect by selected algorithms. Values of RMSE of slope and aspect have been transformed using formula: $y = \ln(x \times 1000)$ for illustration.

- (b) How can we determine an appropriate DEM resolution in relation to slope and aspect computation for a given application?

In a previous study on algorithm error analysis, we have derived the following relationships between error of derived slope and aspect, DEM data error and surface characteristics (Zhou and Liu, 2003):

$$m_S = \sqrt{a^2 M^2 + b^2 m^2} \cos^2 S,$$

$$m_A = \frac{1}{\tan S} \sqrt{a^2 M^2 + b^2 m^2}, \quad (11)$$

where m_S and m_A are the RMSE of derived slope and aspect, respectively, M denotes the algorithm generated error, S denotes slope, m denotes DEM data error (RMSE of DEM), and a and b are the coefficients for M and m , respectively. Note that a and b vary with algorithms as shown in Table 1.

Examining Eq. (11) and Table 1, the overall errors of derived slope and aspect come from three sources, namely, algorithm errors (M) caused by approximation and sampling of continuous surfaces, DEM data error

Table 1

Coefficients a and b for selected slope and aspect algorithms

Algorithm	Coefficient a of M	Coefficient b of m
2FD	$\frac{1}{6} g^2$	$\frac{1}{\sqrt{2}g}$
3FD	$\frac{1}{6} g^2$	$\frac{1}{\sqrt{6}g}$
3FDWRSR	$\frac{1}{6} g^2$	$\frac{1}{\sqrt{5.33}g}$
3FDWRD	$\frac{1}{6} g^2$	$\frac{1}{\sqrt{5.83}g}$
FFD	$\frac{1}{6} g^2$	$\frac{1}{2g}$
SimpleD	$\frac{1}{2} g^2$	$2 \frac{1}{g}$

(m) caused by DEM data capture and generation, and DEM spatial resolution (g , i.e. grid cell size). When DEM resolution tends to zero (i.e. $g \rightarrow 0$), $a \rightarrow 0$, so that the algorithm error will also tend to zero, while the influence of DEM error (m) will tend to infinity ($+\infty$).

Table 2

Computed DEM resolution using RMSE of DEM data and average slope

m (m)	Test surface	RMSE of slope (deg)	Mean slope (deg)	Computed DEM resolution (m)
0.6	Ellipsoid	1.0	37.0	9.0
	Gauss surface	1.4	1.1	9.9
2.1	Ellipsoid	3.2	37.0	9.8
	Gauss surface	6.1	1.1	8.2
6.6	Ellipsoid	9.2	37.0	10.7
	Gauss surface	19.4	1.1	8.0
9.5	Ellipsoid	12.4	37.0	11.4
	Gauss surface	26.4	1.1	8.5
15.1	Ellipsoid	18.0	37.0	12.5
	Gauss surface	37.4	1.1	9.5
19.5	Ellipsoid	21.3	37.0	13.6
	Gauss surface	44.0	1.1	10.4

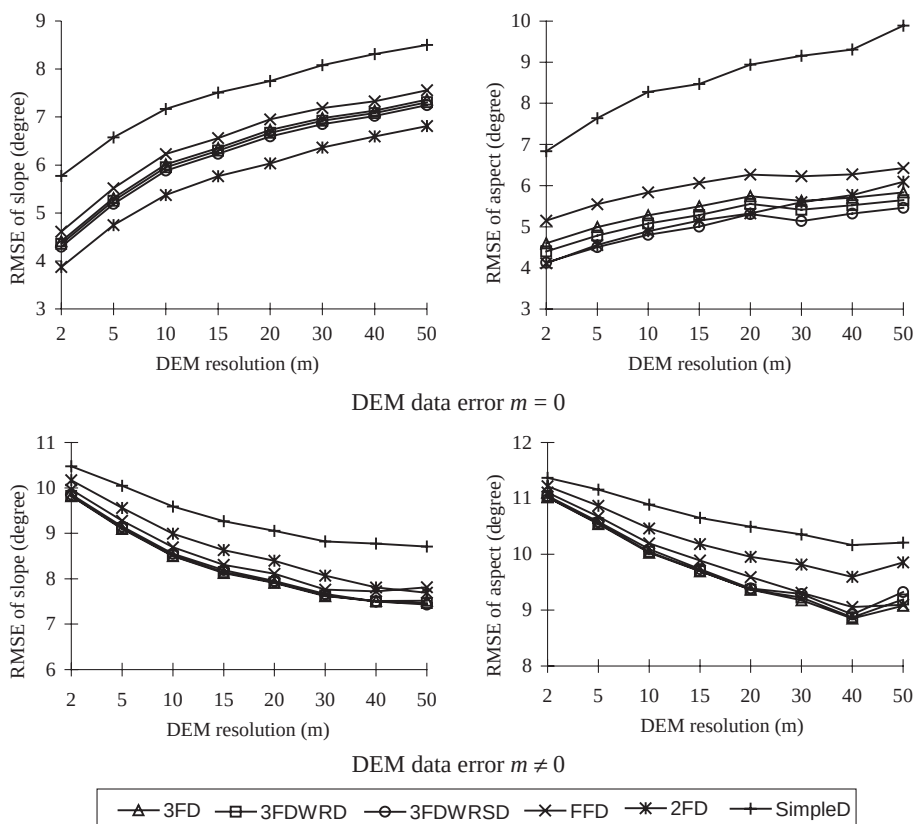


Fig. 6. Relationship between DEM resolution and RMSE of derived slope and aspect on an ellipsoid surface. Values of RMSE of slope and aspect have been transformed using formula: $y = \ln(x \times 1000)$ for illustration.

In other words, for slope and aspect computation, the impact of algorithm error is positively proportional to DEM resolution, while the influence of DEM error is

negatively proportional to DEM resolution. With a higher DEM resolution, the level of detail is increasing (i.e. the surface is better represented), but the influence

of data error is also increased at the same time. With a lower DEM resolution, the impact of data error is decreased, but algorithm errors will cause more significant error on the derived results.

When ignoring algorithm error M , we can determine an appropriate DEM resolution according to known DEM error (m) and average slope (S) of the region:

$$g = \frac{bm}{m_S} \times \frac{180^\circ}{\pi} \cos^2 S. \quad (12)$$

Table 2 illustrates the computed DEM resolution for DEM with various error levels (specified by m —RMSE of DEM data).

While considering the influence of DEM resolution on selected algorithms, we have conducted tests to analyse the relationships of slope and aspect errors and DEM resolution with or without DEM data error. Figs. 6 and 7 show the test results on the ellipsoid and Gauss synthetic surface, respectively.

As shown by Figs. 6 and 7, when DEM data error is minimal, RMSE of slope and aspect increases with lower resolution (i.e. larger grid cell size), regardless of which algorithm is used. The RMSE of derived slope and aspect is positively proportional to grid resolution. When DEM data error is significant, the RMSE of derived slope and aspect is decreasing with lower

resolution, showing a negative proportional relationship to the DEM resolution.

The test results confirm the relationship described by Eq. (11). We therefore conclude that high-resolution DEM does not assure higher slope and aspect accuracy. The better results may only be possible with high DEM data accuracy. In reality, DEM data often contain significant level of errors. It is therefore argued that a higher-resolution DEM does not lead to higher accuracy of estimated slope and aspect. Rather, the accuracy of derived slope and aspect is increasing with lower DEM resolution.

4.3. Grid orientation

At any given point on a surface, slope and aspect are constant parameters, which do not change with grid orientation. However, as DEM organises ground elevation using regularly distributed sample points, different grid orientation may result in errors in computing partial derivatives for slope and aspect computation.

In order to analyse the influence of grid orientation on derived slope and aspect, we have rotated the ellipsoid and Gauss synthetic surface with an increment of 15° to establish DEMs at directions of $15^\circ, 30^\circ, \dots$, and 345° . When the function defining the surface is known, the 'true' slope and aspect value at a given point on the

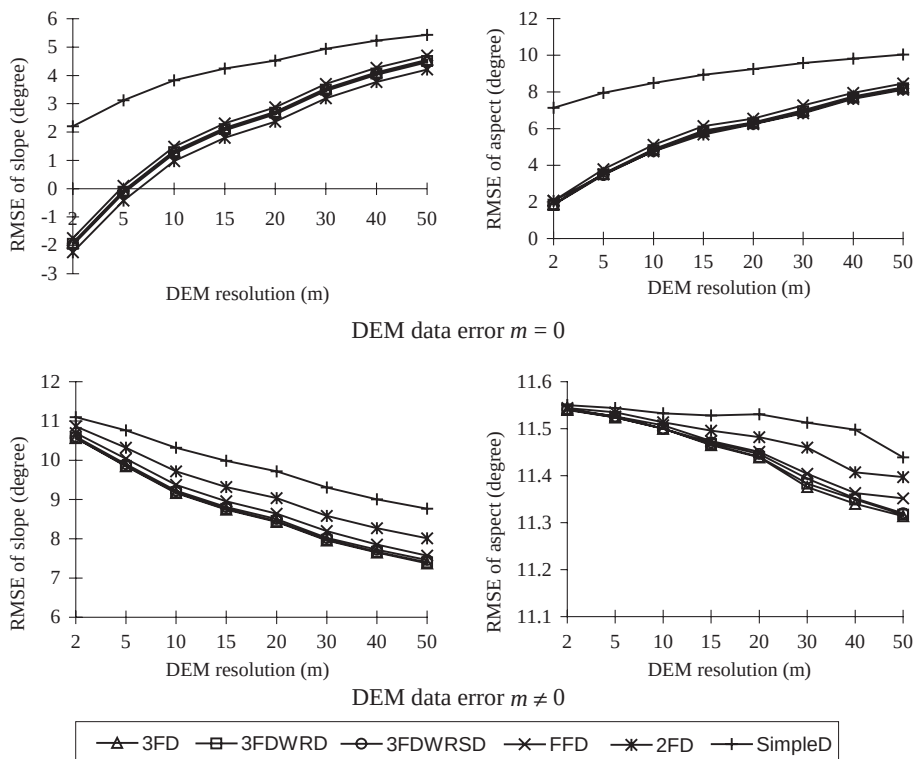


Fig. 7. Relationship between DEM resolution and RMSE of derived slope and aspect on a Gauss synthetic surface. Values of RMSE of slope and aspect have been transformed using formula: $y = \ln(x \times 1000)$ for illustration.

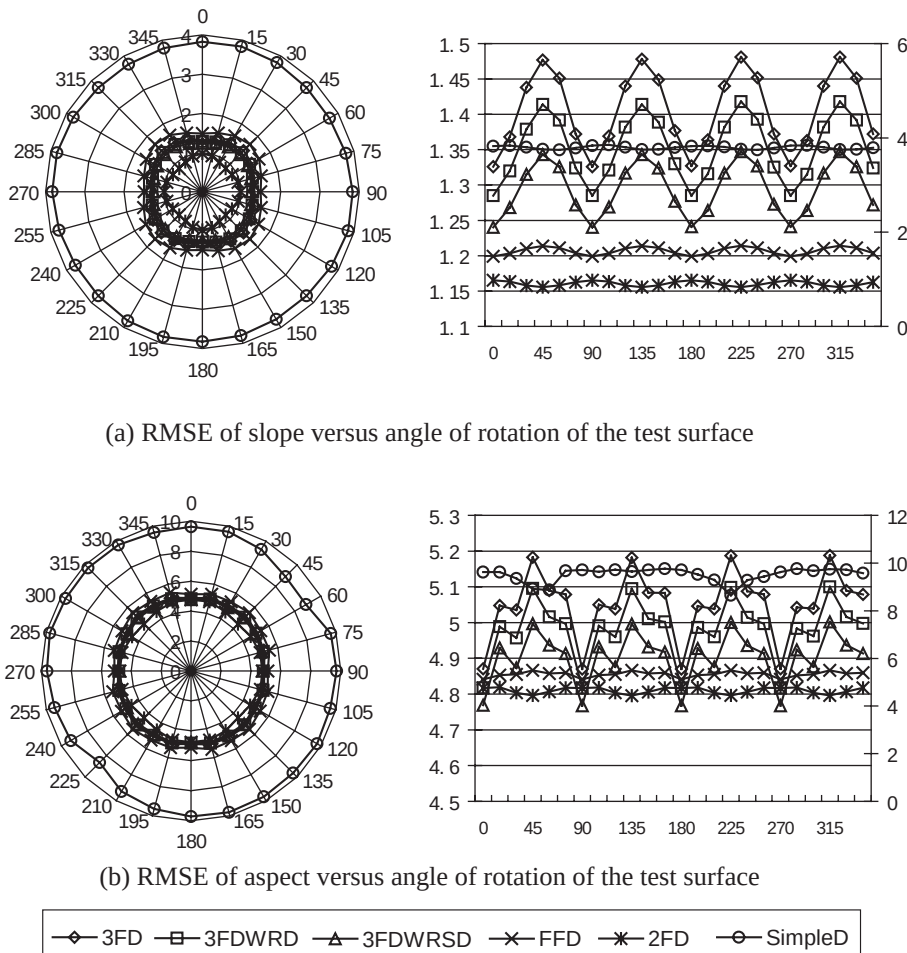


Fig. 8. Influence of grid orientation on slope and aspect results on an ellipsoid surface. Values of RMSE of slope and aspect have been transformed using formula: $y = \ln(x \times 1000)$ for illustration. Left vertical axis shows RMSE (°) for 3FD, 3FDWRD and 3FDWRSR, and right axis shows RMSE (°) for 2FD, FFD and SimpleD.

surface can be computed. The selected algorithms were then applied on these rotated surfaces to derive the results as shown by Figs. 8 and 9 for the ellipsoid and Gauss synthetic surface, respectively.

Observing Figs. 8 and 9 reveals the following findings:

- DEM orientation has greater influence on third-order finite difference algorithms (including 3FD, 3FDWRD and 3FDWRSR) than other algorithms (2FD, FFD and SimpleD). 2FD, FFD and SimpleD have shown almost constant error level in all directions, while 3FD series has shown great changes in association with directions.
- All algorithms show extreme values at $45^\circ \times k$ ($k=0,1,\dots,7$), i.e. demonstrating the octant pattern in directional distribution.
- Errors in slope and aspect synchronise with each other and reach the extreme values with the same

directional pattern. This is because the errors in slope and aspect are related to slope itself, as described by Eq. (11).

5. Conclusion

In this paper we have reported the results of error analysis on derived slope and aspect from DEM by numerous algorithms. The focus of the discussion has been on the influence of DEM data characteristics on the derived slope and aspect parameters and the sensitivity of the selected algorithms in response to the DEM data structure. The impacts of DEM data precision, grid resolution and orientation on derived slope and aspect values has been analysed and tested on artificial surfaces defined by mathematical

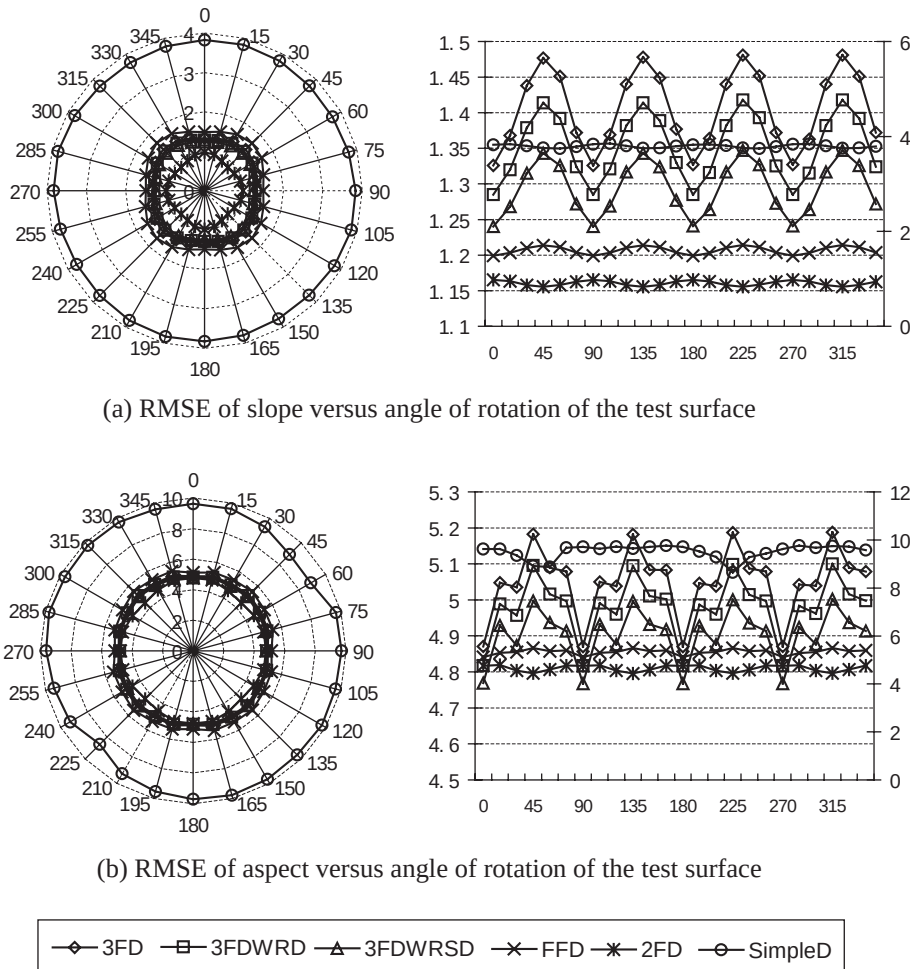


Fig. 9. Influence of grid orientation on slope and aspect results on a Gauss synthetic surface. Values of RMSE of slope and aspect have been transformed using formula: $y = \ln(x \times 1000)$ for illustration. Left vertical axis shows RMSE (°) for 3FD, 3FDWRD and 3FDWRS, and right axis shows RMSE (°) for 2FD, FFD and SimpleD.

functions such as ellipsoid and Gauss synthetic surface.

The findings of this study can be summarized as:

- Algorithm choice is important when data precision is high. When the precision level is reduced, its influence on different algorithms becomes less important. When the error level in a DEM is high, the round-up errors by reducing data precision can be neglected. When DEM data accuracy is higher than the precision, however, the data precision error must be considered while selecting algorithms. Among the selected algorithms, SimpleD seems to produce the worst result at the high precision level, but difference between all other algorithms is quite minimal. The difference among results derived from all algorithms becomes insignificant at the low level of precision.

- A high-resolution DEM does not assure higher slope and aspect accuracy. The better results may only be possible with high DEM data accuracy. In reality where DEM data often contains errors, the accuracy of derived slope and aspect is increasing with lower DEM resolution.
- Grid orientation does cause directional bias on derived slope and aspect, and the 3FD algorithm series has shown the most significant errors due to the change of grid orientation.

This study has shown that using a selection of mathematical surfaces with controlled parameters and data errors, digital terrain analysis (DTA) algorithms can be objectively compared and evaluated, independently from data and human analyst's bias. It is also shown that impact of individual factors can be

independently examined by this approach so that appropriate justification can be made according to the application requirements and data characteristics.

Further studies will be focused on analysing the impact of DEM data errors and surface characteristics on terrain analysis results, such as slope profile and curvature, catchment areas, drainage networks and other derived geomorphic parameters. The real-world tests will also be conducted to compare with the findings by the theoretical analysis. Based on these analysis, the ultimate goal is to set a conclusive guideline for deriving geomorphic parameters from DEM for a given application project.

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References

- Brasington, J., Richards, K., 1998. Interactions between model predictions, parameters and DTM scales for TOPMODEL. *Computers & Geosciences* 24 (4), 299–314.
- Carter, J., 1992. The effect of data precision on the calculation of slope and aspect using gridded DEMs. *Cartographica* 29 (1), 22–34.
- Chang, K., Tsai, B., 1991. The effect of DEM resolution on slope and aspect mapping. *Cartography and Geographic Information Systems* 18, 69–77.
- Chu, T.H., Tsai, T.H., 1995. Comparison of accuracy and algorithms of slope and aspect measures from DEM. In: *Proceedings of GIS AM/FM ASIA'95*. 21–24 August, Bangkok, pp. 1-1-1–1-1-11.
- Davis, F.W., Dozier, J., 1990. Information analysis of a spatial database for ecological land classification. *Photogrammetric Engineering and Remote Sensing* 56, 605–613.
- Fleming, M.D., Hoffer, R.M., 1979. Machine processing of Landsat MSS data and LARS Technical Report 062879. Laboratory for Applications of Remote Sensing, Purdue University, West Lafayette, IN, USA.
- Florinsky, I.V., 1998a. Accuracy of local topographic variables derived from digital elevation models. *International Journal of Geographical Information Science* 12 (1), 47–61.
- Florinsky, I.V., 1998b. Combined analysis of digital terrain models and remotely sensed data in landscape investigations. *Progress in Physical Geography* 22 (1), 33–60.
- Florinsky, I.V., Kuryakova, G.A., 2000. Determination of grid size for digital terrain modeling in landscape investigations—exemplified by soil moisture distribution at a micro-scale. *International Journal of Geographical Information Science* 14 (8), 815–832.
- Gao, J., 1998. Impact of sampling intervals on the reliability of topographic variables mapped from grid DEMs at a micro-scale. *International Journal of Geographical Information Science* 12 (8), 875–890.
- Garbrecht, J., Martz, L.W., 1994. Grid size dependency of parameters from digital elevation models. *Computer and Geosciences* 20 (1), 85–87.
- Hodgson, M.E., 1995. What cell size does the computed slope/aspect angle represent? *Photogrammetric Engineering and Remote Sensing* 61, 513–517.
- Horn, B.K.P., 1981. Hill shading and the reflectance map. *Proceedings of IEEE* 69 (1), 14–47.
- Jones, K.H., 1998. A comparison of algorithms used to compute hill slope as a property of the DEM. *Computers & Geosciences* 24 (4), 315–323.
- Li, Z., Zhu, Q., 2000. *Digital Elevation Models*. Wuhan Science and Technology University of Surveying and Mapping Press, Wuhan, China (in Chinese).
- Liu, X., 2002. On the accuracy of the algorithms for interpreting grid-based digital terrain model. Ph.D. Dissertation, Wuhan University, Wuhan, China (in Chinese).
- Moore, I.D., Lewis, W., Gallant, J.C., 1993. Terrain attribute: estimation methods and scale effects. In: Jakeman, A.J., Beck, M.B., McAleer, M.J. (Eds.), *Modelling Change in Environmental Systems*. Wiley, New York, 189–214.
- O'Callaghan, J.F., Mark, D.M., 1984. The extraction of drainage networks from digital elevation data. *Computer Vision, Graphics, and Image Processing* 28, 323–344.
- Ritter, D., 1987. A vector-based slope and aspect generation algorithm. *Photogrammetric Engineering and Remote Sensing* 53 (8), 1109–1111.
- Sharpnack, D.A., Akin, G., 1969. An algorithm for computing slope and aspect from elevations. *Photogrammetric Survey* 35, 247–248.
- Skidmore, A.K., 1989. A comparison of techniques for calculation gradient and aspect from a gridded digital elevation model. *International Journal of Geographical Information Sciences* 4 (1), 33–49.
- Tang, G., 2000. *A Research on the Accuracy of Digital Elevation Models*. Science Press, Beijing.
- Theobald, D.M., 1989. Accuracy and bias issues in surface representation. In: Goodchild, M.F., Gopal, S. (Eds.), *The Accuracy of Spatial Database*. Taylor and Francis, New York, pp. 99–106.
- Unwin, D., 1981. *Introductory Spatial Analysis*. Methuen, London and New York.
- Wood, J.D., 1996. The geomorphological characterisation of digital elevation model. Ph.D. Thesis, University of Leicester.
- Zevenbergen, L.W., Thorne, C.R., 1987. Quantitative analysis of land surface topography. *Earth Surface Processes and Landforms* 12, 47–56.
- Zhou, Q., Liu, X., 2002. Error assessment of grid-based flow routing algorithms used in hydrological models. *International Journal of Geographical Information Science* 16 (8), 819–842.
- Zhou, Q., Liu, X., 2003. The accuracy assessment on algorithms that derive slope and aspect from DEM. In: Shi, W., Goodchild, M.F., Fisher, P.F. (Eds.), *Proceedings of the Second International Symposium on Spatial Data Quality*. 19–20 March, Hong Kong, pp. 275–285.